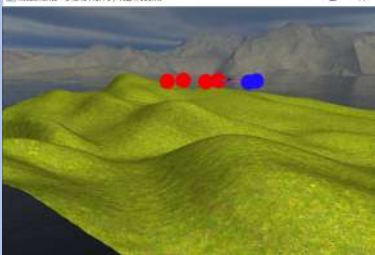
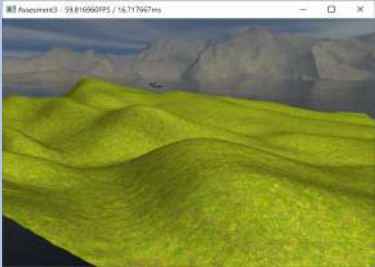


Introduction

<i>Please explain why you implemented this scene</i>	<i>Describe your inspirations</i>	<i>Provide a general description of the scene.</i>
It gave a good opportunity for nice animations, open lighting from the sun and the ability to mess around with generating heightmap terrain.	<i>I'm a fan of planes so thought it would be nice to have something.</i>	<i>Grassy hills with a jet flying above and some scattered trees.</i>

TR 2 – 3D Modelling

Object 1 - procedurally generated

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of object</i>
 	<i>Terrain.cpp</i> <i>CompleteObject.cpp</i> - <i>`constructHeightMapTerrain`</i> <i>line 9</i> <i>110 (flat plane but unused)</i>	Rolling grassy hills generated using a heightmap. Uses indices to save on VRam

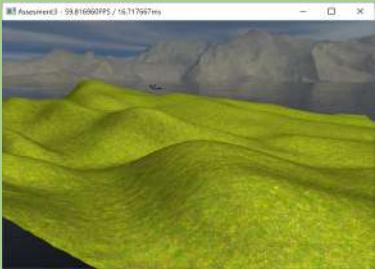
Object 2 - OBJ parser

<i>Please give a screenshot</i>	<i>Provide the URL for the OBJ file(s) you submitted</i>	<i>Describe what you did to extend your OBJ parser from CW2.</i>
	https://sketchfab.com/3d-models/f-22-raptor-update-d-fighter-jet-free-7bfc05d3916b454da4960fc17c093874 <i>Tree from CW2</i> <i>\Assessment3\Assessment3\Assessment3\objs</i>	Normal Coordinates un-reversed UV coordinates (flip with stbi instead) Ns, Ka, Kd and Ks for each material to allow for specific lighting. Modified to use std::optional and std::string to be nicer to use.

		
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TR 3 – 3D Transformations

Object 1 - procedurally generated

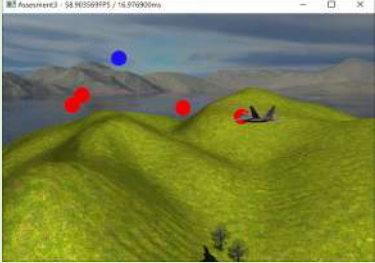
<i>Please give a screenshot of transformed object</i>	<i>reference specific code (filename and line)</i>	<i>Description of transformations</i>
	<i>Terrain.cpp - line28</i>	<i>A y shift is used to move the terrain up to be centred on 0,0 more.</i>

Object 2 - OBJ parser

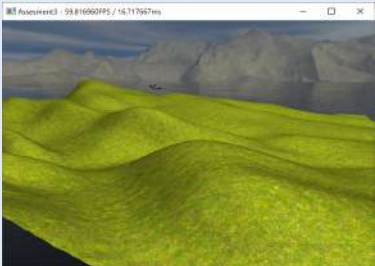
<i>Please give a screenshot of transformed object</i>	<i>reference specific code (filename and line)</i>	<i>Description of transformations</i>
	<i>Assessment3.cpp 156 - 169</i> <i>CompleteObject.cpp 76, 80, 21</i>	<i>Plane animated using translate + rotate to follow a curve.</i> <i>Plane + trees scaled to fit the scene size. Trees transformed to be placed correctly.</i>

TR 3 – Animation

<i>Please give a screenshot of animated object</i>	<i>reference specific code (filename and line)</i>	<i>Description of animation</i>
	<i>F22.cpp - Full file (specifics on line 24)</i> <i>CompleteObject - 21.</i>	<i>Plane flies around interpolated bezier curve to go around an oval in 3D space and follow the tangent of the curve.</i>

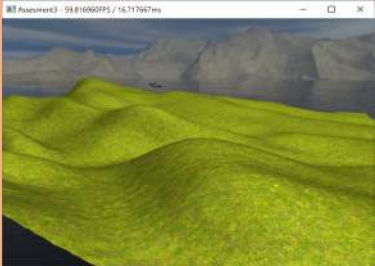
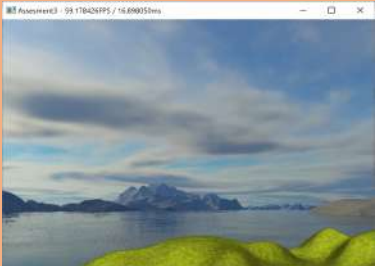
		
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TR 4 – Camera

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of camera</i>
	<code>camera.cpp</code>	Flythrough Camera

TR 5 – Texture

Object 1 - procedurally generated

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of texture</i>
 	<code>Texture.cpp</code>, - 3 <code>mipMaps</code> - 37, <code>Terrain.cpp</code> 47 <code>Skybox.frag</code> .vert and .cpp <code>Texture.cpp</code> 47	Uses mipmaps UV's calculated to set resolution to tessellate seamlessly and still appear high res. Skybox uses cube mapping.

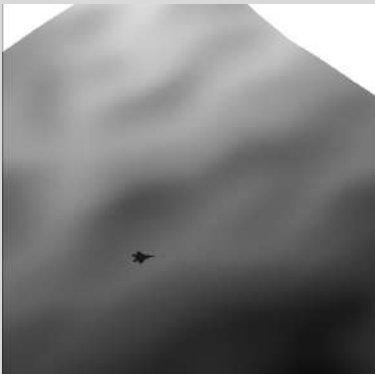
TR 6 – Lighting

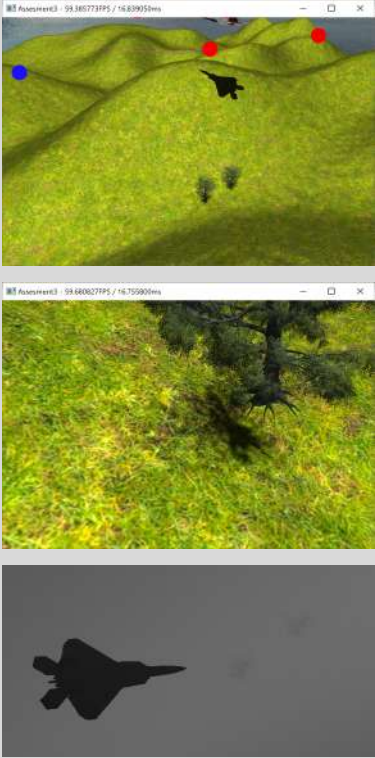
Object 1 - procedurally generated

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of light(s) on object</i>
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	Phong.frag, vert	Blinn-phong lighting Directional (spot and positional are implemented but un-used)
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Object 2 - OBJ parser		
Please give a screenshot	reference specific code (filename and line)	Description of light(s) on object
	Phong.frag, vert	Blinn-phong lighting Directional spot and positional are implemented but un-used)

TR 7 - Shadow		
Object 1 - procedurally generated		
Please give a screenshot	reference specific code (filename and line)	Description of shadow on object
	Phong.frag -162 ShadowRendering.cpp shadow.cpp	Shadows use PCF to increase resolution.
Object 2 - OBJ parser		

Please give a screenshot	reference specific code (filename and line)	Description of shadow on object
	Phong.frag -162 ShadowRendering.cpp shadow.cpp shadow.frag,vert	Shadows use PCF to increase resolution. Pass through texture transparency to improve shadow quality for tree.

TR 8 - Interactive object

Please give a screenshot	reference specific code (filename and line)	Description of interactive object
NAN	NAN	NAN

TR 9 – Curves

Please give a screenshot	reference specific code (filename and line)	Description of curve(s)
	Castelijau.cpp (wholeFile) F22.cpp (wholeFile)	Two bezier curves in a 3D plane used for object animation. Tangent of line is calculated to apply rotation to the animated object.

TR 10 – Transparency

Please give a screenshot	reference specific code (filename and line)	Description of transparency
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	Phong.frag 43,44	If transparency is present in a texel we drop the fragment which uses that.
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R&D

Please provide details of any research and development you conducted, as additional technique(s) not in the lecture notes.

Please give a screenshot	reference specific code (filename and line)	Description of Research including websites, articles, references, etc.
	Phong.frag 55 - 63	Blinn-phong lighting https://learnopengl.com/Advanced-Lighting/Advanced-Lighting
	Terrain.cpp 89, 9 texture.cpp 3, 79	Heightmaps https://learnopengl.com/Guest-Articles/2021/Tessellation/Height-map
	Skybox.cpp texture.cpp 47 F22.cpp, castelijai.cpp, 34	Skybox + cubemap Bezier curve tangents for use as animation https://stackoverflow.com/questions/25453159/getting-consistent-normals-from-a-3d-cubic-bezier-path
	Phong.frag -162	PCF (improve shadow quality) https://learnopengl.com/Advanced-Lighting/Shadows/Shadow-Mapping
	Assessment3.cpp 280 - 282, obj.cpp 28	Alternate light properties https://learnopengl.com/Lighting/Materials
	Terrain.cpp - 50	Indices https://learnopengl.com/Guest-Articles/2021/Tessellation/Height-map